

OpenCode + Ollama + Qwen3-Coder 30B

Complete Linux Mint installation and GPU configuration guide

	Laptop	Desktop PC
CPU	Intel Core i5 10th Gen	Intel Core i5-12400F
RAM	24 GB	64 GB
GPU	NVIDIA MX230	NVIDIA RTX 3070 Ti
Typical VRAM	Usually 2 GB	Usually 8 GB
30B model	Primarily CPU/system RAM; slow	GPU + system RAM; preferred
Starting context	8K–16K	16K–32K, then increase if stable

1. Important: no dataset is required

Qwen3-Coder is already trained. You do not download or prepare a training dataset. The roughly 19 GB Ollama download is the quantized model weights. OpenCode supplies relevant project files and context while you work.

Hardware warning: the 30B Q4_K_M model is roughly 19 GB. Your laptop has only 24 GB RAM, so it has very little headroom for Linux, OpenCode, Ollama and context. The desktop is substantially better suited to this model.

2. Install and verify Ollama

```
sudo apt update && sudo apt upgrade -y
sudo reboot

# after reboot
curl -fsSL https://ollama.com/install.sh | sh
ollama --version
systemctl status ollama
sudo systemctl enable --now ollama
```

Verify the local API:

```
ollama list
curl http://localhost:11434/api/tags
```

3. Check NVIDIA graphics support BEFORE downloading the model

For both machines, check whether Linux sees the NVIDIA GPU:

```
nvidia-smi
```

If this command shows the GPU name, driver version and VRAM, the proprietary NVIDIA driver is working. If it says the command is missing or cannot communicate with the driver, fix the NVIDIA driver first. You do **not** normally need to install the CUDA Toolkit separately just to make Ollama use an NVIDIA GPU.

4. Laptop GPU configuration — MX230

The MX230 is the limiting component. It normally has only about 2 GB VRAM, far less than the 19 GB Qwen3-Coder model. Do not try to force the whole model into the MX230. Ollama can use available GPU acceleration while keeping the remainder in system RAM/CPU.

Check the GPU:

```
nvidia-smi
```

While a model is running, monitor it with:

```
watch -n 1 nvidia-smi
```

For the laptop, expect the 30B model to be predominantly CPU/system-RAM bound. GPU memory may be used for some layers, but the MX230 will not make the 30B model fast. Keep the laptop plugged in and expect significant heat and fan

activity.

Do not install extra CUDA packages just for this. First establish that the normal NVIDIA driver works with nvidia-smi; Ollama handles supported NVIDIA acceleration.

5. Desktop GPU configuration — RTX 3070 Ti

The RTX 3070 Ti desktop is much better suited to Qwen3-Coder 30B. A typical 3070 Ti has 8 GB VRAM, while the model is about 19 GB, so the complete model cannot fit in VRAM. Ollama can therefore use a hybrid CPU/GPU arrangement: some work is accelerated by the GPU while the remaining model state is kept in system RAM.

First verify the driver:

```
nvidia-smi
```

Then monitor GPU usage while Qwen is running:

```
watch -n 1 nvidia-smi
```

You do **not** need to manually move the model to VRAM, set CUDA paths, or force the entire model into the GPU. Let Ollama manage the CPU/GPU split.

If NVIDIA's driver is not working, use Linux Mint's Driver Manager to install the recommended proprietary NVIDIA driver for your Mint release, reboot, and run `nvidia-smi` again.

6. Download Qwen3-Coder 30B

```
ollama pull qwen3-coder:30b
```

The current Ollama Qwen3-Coder 30B Q4_K_M download is approximately 19 GB. Keep at least 30–40 GB of free disk space available.

```
ollama list
```

7. Test the model

```
ollama run qwen3-coder:30b
```

Try: **Write a Python program that calculates the Fibonacci sequence.** Exit with:

```
/bye
```

8. Monitor RAM, model and GPU

```
ollama ps  
free -h  
htop  
nvidia-smi
```

For continuous GPU monitoring:

```
watch -n 1 nvidia-smi
```

Use `ollama ps` to inspect the currently loaded model. On the laptop, watch RAM closely; on the desktop, watch both VRAM and system RAM.

9. Install OpenCode

```
curl -fsSL https://opencode.ai/install | bash
opencode --version
```

Connect OpenCode to local Ollama using:

```
ollama launch opencode --model qwen3-coder:30b
```

10. Use OpenCode with a project

```
cd ~/Projects/my-project
ollama launch opencode --model qwen3-coder:30b
```

Example tasks:

- Analyze this project and explain its architecture.
- Find and fix the bug causing the application to crash when the user logs in.
- Add dark mode to this application.

11. Laptop: reduce context to protect 24 GB RAM

Do not begin with the model's maximum 256K context on a 24 GB laptop. Start at 16K; if memory pressure or swapping occurs, use 8K.

```
nano ~/qwen3-coder.Modelfile

FROM qwen3-coder:30b

PARAMETER num_ctx 16384

# save: Ctrl+O, Enter
# exit: Ctrl+X

ollama create qwen3-coder-local -f ~/qwen3-coder.Modelfile
ollama run qwen3-coder-local
ollama launch opencode --model qwen3-coder-local
```

If 16K still causes heavy swapping, change 16384 to 8192, recreate the model, and test again.

12. Desktop: start moderate, then tune

With 64 GB RAM and an RTX 3070 Ti, start around 16K–32K context. If performance and memory usage are comfortable, you can experiment with a larger context. Larger context is not free: it increases memory requirements.

Do not assume that the advertised 256K context is practical on every machine. The maximum supported context and the practical context you can run smoothly are different things.

13. Complete installation sequence

Both machines:

```
sudo apt update && sudo apt upgrade -y
curl -fsSL https://ollama.com/install.sh | sh
sudo systemctl enable --now ollama

nvidia-smi

ollama pull qwen3-coder:30b
ollama run qwen3-coder:30b

curl -fsSL https://opencode.ai/install | bash
opencode --version

ollama launch opencode --model qwen3-coder:30b
```

14. Laptop-specific final setup

```
# Check NVIDIA driver
```

```

nvidia-smi

# Model
ollama pull qwen3-coder:30b

# Create 16K-context variant
nano ~/qwen3-coder.Modelfile

FROM qwen3-coder:30b
PARAMETER num_ctx 16384

ollama create qwen3-coder-local -f ~/qwen3-coder.Modelfile

# Launch
ollama launch opencode --model qwen3-coder-local

```

Expected behavior: mostly CPU/system-RAM inference, potentially slow. If the machine swaps heavily, lower context to 8K.

15. Desktop-specific final setup

```

# Verify NVIDIA
nvidia-smi

# Install model
ollama pull qwen3-coder:30b

# Test
ollama run qwen3-coder:30b

# Launch OpenCode
ollama launch opencode --model qwen3-coder:30b

# Monitor in another terminal
watch -n 1 nvidia-smi
ollama ps
free -h

```

Expected behavior: Ollama can use the RTX 3070 Ti for available GPU acceleration and system RAM for the remainder. The entire ~19 GB model does not fit into a typical 8 GB 3070 Ti VRAM allocation.

16. Useful commands

Purpose	Command
List models	<code>ollama list</code>
Run model	<code>ollama run qwen3-coder:30b</code>
Show loaded model	<code>ollama ps</code>
Remove model	<code>ollama rm qwen3-coder:30b</code>
Download/update	<code>ollama pull qwen3-coder:30b</code>
Check service	<code>systemctl status ollama</code>
Check NVIDIA	<code>nvidia-smi</code>
Live GPU monitoring	<code>watch -n 1 nvidia-smi</code>
RAM monitoring	<code>free -h</code>

17. Troubleshooting

- **nvidia-smi fails:** install/fix the proprietary NVIDIA driver using Linux Mint's Driver Manager, reboot, and test again.
- **Ollama is not running:** use `sudo systemctl enable --now ollama`.
- **Model download fails:** retry `ollama pull qwen3-coder:30b` and make sure there is sufficient disk space.
- **Laptop freezes or swaps:** reduce context from 16K to 8K and close other memory-heavy applications.
- **Desktop VRAM fills:** this is expected with an 8 GB GPU and a ~19 GB model. Do not try to force the whole model into VRAM.
- **Model is slow:** the laptop's MX230 is not powerful enough to make a 30B model fast. The desktop is the preferred machine.
- **Do not manually edit/delete Ollama model files:** manage models through `ollama list`, `ollama pull`, and `ollama rm`.

18. Hardware recommendation summary

Laptop — 24 GB RAM + i5 10th Gen + MX230: Qwen3-Coder 30B can be attempted, but it is a demanding configuration. Expect CPU-heavy, slow inference. Use 8K–16K context and monitor RAM.

PC — 64 GB RAM + i5-12400F + RTX 3070 Ti: this is the machine to use for Qwen3-Coder 30B. The 8 GB-class GPU cannot hold the entire ~19 GB model, but Ollama can use GPU acceleration alongside system RAM. Start at a moderate context and tune upward.